Semiconductors

solids → conductors, semiconductor, Insulator

semiconductor → Intrinsic, Extrinsic

Extrinsic → P type, n type

Conductor : temp ↑ , Resistance ↑ (+ve coef. of R)
Semiconductor : temp ↑ , Resistance ↓ (−ve coef of R)

Intrinsic Semiconductor

When conductivity is only due to breaking of covalent bonds of e^- in valence band.

The Abs. of e^- in VB is leads to formation of hole.

Here, $n = p = n_i$

n_i = no. of charge carriers.

Carrier Concentration And fermi Level.

~~density of state func. g(E)dE~~

Total no. of e^- / unit volume

$$n_e = \int_{E_c}^{top} f(E)\, g(E)\, dE$$

$$= \int_{E_c}^{\infty} f(E) g(E) dE$$

f(E) : fermi-Dirac Distr. func

g(E) : density of state func.

$$f(E) = \frac{1}{e^{(E-E_f)/KT} + 1}$$

$$g(E) = \frac{4\pi}{h^3}(2m_e)^{3/2}(E-E_c)^{1/2}.$$

Ans.

$$n_e = \int_{E_c}^{\infty} \frac{1}{e^{(E-E_f)/k_BT}+1} \times \frac{4\pi}{h^3}(2m_e)^{3/2}(E-E_c)^{1/2}\,dE$$

$\cancel{E \gg E_f}$ $E - E_f \gg k_BT$

$$\therefore n_e = \int_{E_c}^{\infty} \frac{1}{e^{(E-E_f)/KT}} \cdot \frac{4\pi}{h^3}(2m_e)^{3/2}(E-E_c)^{1/2}\,dE$$

$$\frac{4\pi}{h^3}(2m_e)^{3/2} \int_{E_c}^{\infty} \frac{(E-E_c)^{1/2}}{e^{(E-E_f)/KT}}\,dE$$

$$= \frac{4\pi}{h^3}(2m_e)^{3/2} \int_{E_c}^{\infty} (E-E_c)^{1/2} e^{(E_f-E)/KT}\,dE$$

Adding & sub. E_c from the exp. term.

$$e^{(E_f-E+E_c-E_c)/KT} = e^{(E_F-E_c)/KT}\, e^{(E_c-E)/KT}$$

$$\cancel{n_e = \frac{4\pi}{h^3}(2m_e)^{3/2}\int_{E_c}^{\infty}}$$

Let $(E-E_c)/K_BT = x$

$dE = K_BT\,dx$

$(E-E_c)^{1/2} = (x)^{1/2}(K_BT)^{1/2}$

$$n_e = \frac{4\pi}{h^3}(2m_e)^{1/2} e^{(E_f-E_c)KT} \int_{E_c}^{\infty} x^{1/2}(KT)^{1/2}\, e^{x}\,dx$$

$$\boxed{n_e = 2\left(\frac{2\pi m_e KT}{h^2}\right) e^{(E_f-E_c)/KT}}$$

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$$n_h = 2\left(\frac{2\pi m_h kT}{h^2}\right) e^{(E_f - E_c)/kT}$$

$$n_e = 2\left(\frac{2\pi m_e kT}{h^2}\right) e^{(E_v - E_f)/kT}$$

for intrinsic,

$$n_e = n_h$$

$$(m_e)^{3/2} e^{(E_f - E_c)/kT} = (m_h)^{3/2} e^{(E_v - E_f)/kT}$$

$$e^{(2E_f - E_c - E_v)/kT} = \left(\frac{m_h}{m_e}\right)^{3/2}$$

$$\frac{2E_f - E_c - E_v}{kT} = \frac{3}{2}\log\left(\frac{m_h}{m_e}\right)$$

for $m_h = m_e$, $\log\left(\frac{m_h}{m_e}\right) = 0$

$$\therefore \frac{2E_f - E_c - E_v}{kT} = 0$$

$$\boxed{E_f = \frac{E_c + E_v}{2}}$$

Extrinsic Semiconductor

To increase the conductivity, doping of trivalent or pentavalent comp is done.

N type semiconductor

pentavalent Impurity

$e^- >>> $ holes (donor Impurity)

Here, fermi level lies ~~at half way b/w~~ below the conduction band.

at 0 K → $E_f = \frac{E_c + E_d}{2}$

P type semiconductor.

trivalent impurity.

$h >>> e^-$ (Acceptor Impurity)

Here, fermi level lies Above valence Band

Effect of Temp on Fermi-Level

Max energy that the e^- possess at 0K is fermi energy

when $T = 0$, $E > E_f$

$$f(E) = \frac{1}{e^{\infty}+1} = 0$$

when $T = 0$, $E < E_f$

$$f(E) = \frac{1}{e^{-\infty}+1} = 1$$

at $T = T\,k$, at $E = E_f$

$$f(E) = \frac{1}{e^{0}+1} = \frac{1}{2}$$

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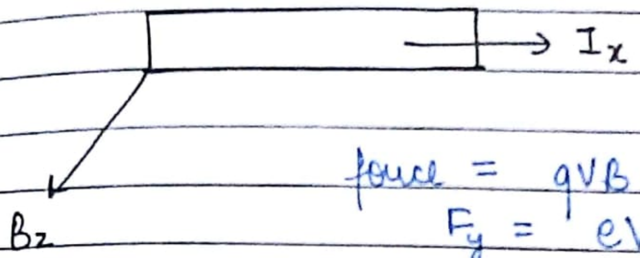
App of Hall's Eff:
Determination of semiconductor type
$R_H \rightarrow + \Rightarrow$ p type
$R_H \rightarrow - \Rightarrow$ n-type..

Hall Effect.

If a piece of conductor carrying current is placed in a transverse magnetic field, then an electric field ~~is~~/ potential diff is generated in a dir. ~~both~~ $\perp$ to both current and magnetic field. This phenomenon is Hall's Effect.



$$\text{force} = qVB$$
$$F_y = eV_x B_z$$
Also $F_y = qE_y = eE_y$

$$\therefore \quad eV_x B_z = eE_y$$
~~Also $B_z = B$~~ $\quad E_y = V_x B_z$ —— (1)

$J_x \rightarrow$ current Density,
$$J_x = \frac{I_x}{A} = nev_x$$

$$V_x = \frac{J_x}{ne}$$

$$E_y = \frac{B_z J_x}{ne} \quad \text{—— (2)}$$

Hall coeff $R_H = \frac{E_y}{J_x B_z}$ —— (3).

$$\therefore \quad R_H = \frac{B_z J_x}{ne J_x B_z}$$

$$\therefore \quad \boxed{R_H = \frac{1}{ne}}$$

for p type: $\boxed{R_H = \frac{1}{pe}}$